

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Concise solution :

- Use the relation between energy and wavelength : $E = hc / \lambda$, so $\lambda = hc / E$.
- With $h = 6.625419996 \times 10^{-27}$ erg $\cdot$ s (Planck's constant in erg $\cdot$ s) and $c = 2.99792458 \times 10^{10}$ cm/s ,
 $\lambda = (6.625 \times 10^{-27}) / (1.0 \times 10^{-3}) = 6.625 \times 10^{-24}$ cm .
- Convert cm to Å : $\lambda = (6.625 \times 10^{-24}$ cm) $\times$ (10 Å/cm) $= 6.625 \times 10^{-23}$ Å $\approx$ 0.06625 nm .
- This corresponds to about 1.23×10^{-7} cm , a typical de Broglie wavelength for electrons with comparable energies .

Correct answer : G . 1.23×10^{-7} cm

</think>

<answer>

G . 1.23×10^{-7} cm

</answer>

